

CPE 381 CLASSWORK 02: MATHEMATICAL PRELIMINARIES

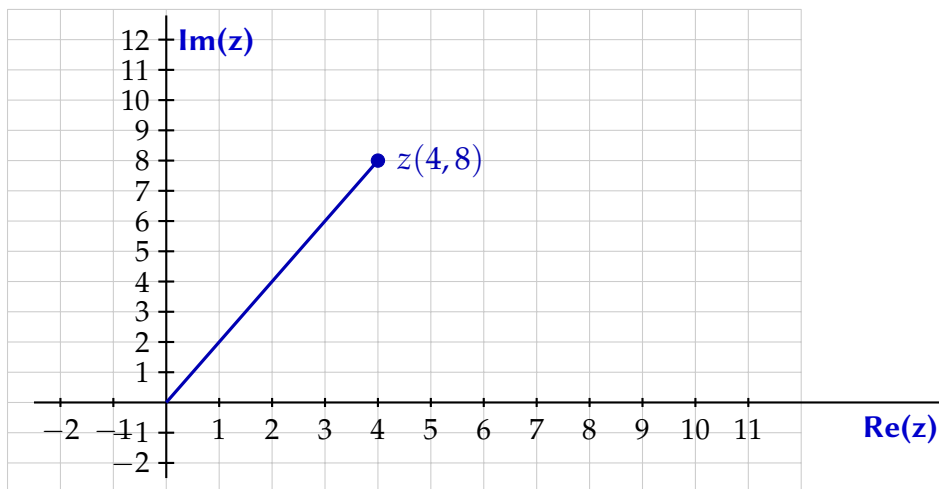
Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes

Due: August 26, 2026 (In-Class)

1. Consider the complex number z on Figure 1. What is the complex conjugate $\bar{z}$? (3 Points)

Figure 1: Complex z .**Answer**

2. If $z = \sqrt{3} + i$, what is z^4 in polar form? Use De Moivre's Theorem. (6 Points)

Answer

3. What is the numerical value of $2(\sin^2 \theta + \cos^2 \theta)$?

(3 Points)

Answer

4. If $\sin(\theta) = \frac{12}{13}$, and θ lies in the second quadrant, find the value of $\sec(\theta) + \tan(\theta)$. **(8 Points)**

Answer